WORKED SOLUTION

STEP III 1992, Question 13

Mechanics • Rotation About a Fixed Axis

STEP III 1992, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q13

The key idea

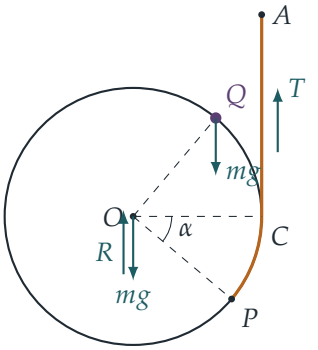
The axis is fixed, so the disc and particle form one rigid body with a single coordinate, the angle turned. Conservation of energy (elastic, gravitational, kinetic) gives the angular speed at every position, and "P reaches C" just means the kinetic energy never falls to zero on the way. Once we have the angular speed at C, the reaction on the axis comes from Newton's second law for the whole system, applied to its centre of mass.

Solution

The model and the geometry

We treat Q as a particle and the string as light. We take the rim as smooth, so the tension T is the same all along the string, and the normal contact forces from the string on the rim pass through O. Hence the string's only moment about O is that of the tension pulling tangentially at P, which is $2bT$. (Exactly the same moment arises if we regard the tension as acting where the string leaves the rim at C: vertically up along AC, at distance $2b$ from O.)

Let $\alpha = \angle POC$, with P below OC. At release $\alpha = \frac{\pi}{2}$, and P reaches C when $\alpha = 0$. Since $\angle POQ = 90°$, OQ makes angle $\frac{\pi}{2} - \alpha$ above OC, so Q is at height $2b\cos\alpha$ above O. In particular Q starts level with O (at C) and is directly above O when P reaches C.



The string's total length is $AC + \text{arc } CP = \pi b + 2b\alpha$, and its natural length is πb, so its extension is $2b\alpha$. By Hooke's law

$$T = \frac{kmg \cdot 2b\alpha}{\pi b} = \frac{2kmg\alpha}{\pi},$$

and the elastic energy stored is

$$E(\alpha) = \frac{kmg\,(2b\alpha)^2}{2\pi b} = \frac{2kmgb\,\alpha^2}{\pi}.$$

The moment of inertia about the axis is that of the disc plus that of Q on the rim:

$$I = \tfrac{1}{2}m(2b)^2 + m(2b)^2 = 6mb^2.$$

P reaches C

As the disc turns so that P rises towards C (so α decreases), the string shortens and gives up elastic energy, while Q rises. The axis is fixed and smooth, so energy is conserved. With the system starting from rest at $\alpha = \frac{\pi}{2}$, where Q is at height 0 relative to O and $E = \frac{1}{2}\pi kmgb$,

$$\begin{aligned}\tfrac{1}{2}I\dot{\alpha}^2 &= E\left(\tfrac{\pi}{2}\right) - E(\alpha) - mg \cdot 2b\cos\alpha \\ &= \frac{2kmgb}{\pi}\left(\frac{\pi^2}{4} - \alpha^2\right) - 2mgb\cos\alpha.\end{aligned}$$

The disc does start to turn the right way: at release the string pulls with $T = kmg$ at radius $2b$, while Q's weight acts at horizontal distance $2b$ the other way, so the net moment is $2(k-1)mgb > 0$, because $k > \frac{4}{\pi} > 1$.

P reaches C provided the kinetic energy stays positive for every α in $[0, \frac{\pi}{2})$, that is, provided

$$\frac{k}{\pi}\left(\frac{\pi^2}{4} - \alpha^2\right) > \cos\alpha \qquad \text{for } 0 \le \alpha < \tfrac{\pi}{2}.$$

Since $k > \frac{4}{\pi}$ and $\frac{\pi^2}{4} - \alpha^2 > 0$, the left-hand side is greater than

$$\frac{4}{\pi^2}\left(\frac{\pi^2}{4} - \alpha^2\right) = 1 - \frac{4\alpha^2}{\pi^2},$$

so it is enough to show that $g(\alpha) = 1 - \dfrac{4\alpha^2}{\pi^2} - \cos\alpha \geq 0$ on $[0, \frac{\pi}{2}]$.

Ask yourself How can we compare a quadratic with $\cos\alpha$ on a whole interval?

Look at the second derivative. We have $g(0) = g(\frac{\pi}{2}) = 0$, $g'(\alpha) = -\frac{8\alpha}{\pi^2} + \sin\alpha$, so $g'(0) = 0$, and $g''(\alpha) = \cos\alpha - \frac{8}{\pi^2}$. Since $\frac{8}{\pi^2} < 1$ and $\cos\alpha$ decreases from 1 to 0, there is a single α_0 with $g'' > 0$ on $[0, \alpha_0)$ and $g'' < 0$ on $(\alpha_0, \frac{\pi}{2}]$.

- On $[0, \alpha_0]$, g' is increasing from $g'(0) = 0$, so $g' \geq 0$ and $g \geq g(0) = 0$.
- On $[\alpha_0, \frac{\pi}{2}]$, g is concave, so its graph lies above the chord joining its endpoint values $g(\alpha_0) \geq 0$ and $g(\frac{\pi}{2}) = 0$; hence $g \geq 0$ there too.

So $g \geq 0$ throughout, and the strict inequality $k > \frac{4}{\pi}$ makes the kinetic energy strictly positive for $0 \leq \alpha < \frac{\pi}{2}$. The disc never comes to rest before $\alpha = 0$, so P reaches C. In particular, at C (put $\alpha = 0$),

$$3mb^2\dot{\alpha}^2 = \frac{\pi k m g b}{2} - 2mgb \quad \Longrightarrow \quad \omega^2 = \frac{(\pi k - 4)g}{6b},$$

where ω is the angular speed when P is at C. (Here we see where the condition $k > \frac{4}{\pi}$ comes from: it is exactly what makes $\omega^2 > 0$.)

The reaction on the axis

When P is at C, two things happen at once.

- The string is at its natural length ($AC = \pi b$ and the arc is gone), so $T = 0$.
- Q is vertically above O, so its weight has no moment about O.

So there is no moment about the axis at that instant, and the angular acceleration is zero. The disc's centre O is fixed, and Q moves on a circle of radius $2b$ with zero angular acceleration, so its acceleration is purely centripetal: $2b\omega^2$ directed from Q towards O, that is, vertically downwards.

Ask yourself Which forces go into the equation for the reaction?

Apply Newton's second law vertically to the whole system (disc plus Q). The external forces are the two weights $2mg$ downwards, the reaction R upwards on the axis, and the string tension, which is zero at this instant. The internal force between Q and the disc cancels. Only Q accelerates, so

$$R - 2mg = m\,(-2b\omega^2),$$

and therefore

$$R = 2mg - 2mb \cdot \frac{(\pi k - 4)g}{6b} = 2mg - \frac{mg(\pi k - 4)}{3} = \frac{mg(10 - \pi k)}{3},$$

as required. (The horizontal component of the reaction is zero at this instant, since Q has no horizontal acceleration.)

Checks

- At the borderline $k = \frac{4}{\pi}$ the system would just creep up to C with $\omega = 0$, and the formula gives $R = 2mg$, the static value. That is consistent.
- A computer integrates the equation of motion $6mb^2\ddot{\psi} = 2bT - 2mgb\cos\psi$ (with $\psi = \frac{\pi}{2} - \alpha$) numerically. With $m = g = b = 1$ and $k = 2$ it reaches C with $\omega^2 = 0.38053$, matching $(2\pi - 4)/6$, and $R = 1.23894 = (10 - 2\pi)/3$. It agrees to six decimal places for $k = 1.29$, 1.5, 3 as well.
- If $k > \frac{10}{\pi}$ the formula gives $R < 0$: the axis must pull down, because Q is whirling over the top fast enough to need more than the system's weight as centripetal force.

Common slips

- Taking the extension as πb throughout. In fact it shrinks to zero as P reaches C, which is why the tension vanishes there.
- Forgetting the disc's own moment of inertia $\frac{1}{2}m(2b)^2$ and using only Q's.
- Showing only that the kinetic energy at C is positive. "P reaches C" needs the kinetic energy to stay positive all the way there.

Beyond STEP

The final step is the rigid-body version of "total external force equals total mass times acceleration of the centre of mass", and the reaction at a pivot is a classic application of it. A first-year rigid-body dynamics course develops this together with the angular-momentum equation $\mathrm{d}\mathbf{L}/\mathrm{d}t = \boldsymbol{\tau}$. The inequality $\cos\alpha \le 1 - 4\alpha^2/\pi^2$ is a cousin of Jordan's inequality $\sin x \ge 2x/\pi$, and it was proved here by the standard convexity argument. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats fixed-axis rotation and pivot reactions very clearly.

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